
Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the database design of the OptiCrop Smart Agricultural Production Optimization System. It defines the major entities involved in crop recommendation and shows how they interact to support intelligent agricultural decision-making.

ENTITIES INCLUDED

The ER model contains the following entities:

- User
- SoilData
- Crop
- Dataset
- MLModel
- Prediction
- Report

Each entity contains a unique identifier:

Primary keys

- User: user_id
- SoilData: soil_id
- Crop: crop_id
- MLModel: model_id
- Report: report_id
- Dataset: dataset_id
- Prediction: prediction_id

Foreign keys

- SoilData references User through user_id.
- Prediction references SoilData through soil_id.
- Prediction references Crop through crop_id
- Prediction references MLModel through model_id.
- Report references Prediction through prediction_id.

Relationships

- A user can submit multiple soil records for analysis.
- Each soil record is processed to generate crop recommendations.
- A crop may appear in multiple prediction results.
- One machine learning model can generate many predictions.
- A dataset can be used to train multiple machine learning models.
- Prediction outputs are used to generate recommendation reports.

Database Structure

The ERD separates agricultural data, crop information, prediction records, datasets, machine learning models, and reports into distinct entities. This approach improves maintainability, reduces data duplication, and supports efficient storage and retrieval of information.

The ER model supports:

- Collection and management of soil information.
- Storage of environmental parameters.
- Crop prediction using machine learning techniques.
- Management of training datasets and models.
- Generation of crop recommendation reports.
- Support for data-driven farming decisions.

B ENEFITS

The proposed database design enables scalability, improves data consistency, and helps deliver accurate crop recommendations to farmers based on soil and environmental conditions.

